

Decomposition and correction overlapping peaks of LIBS using an error compensation method combined with curve fitting

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The laser induced breakdown spectroscopy (LIBS) technique is an effective method to detect material composition by obtaining the plasma emission spectrum. The overlapping peaks in the spectrum are a fundamental problem in the qualitative and quantitative analysis of LIBS. Based on a curve fitting method, this paper studies an error compensation method to achieve the decomposition and correction of overlapping peaks. The vital step is that the fitting residual is fed back to the overlapping peaks and performs multiple curve fitting processes to obtain a lower residual result. For the quantitative experiments of Cu, the Cu-Fe overlapping peaks in the range of 321–327 nm obtained from the LIBS spectrum of five different concentrations of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution were decomposed and corrected using curve fitting and error compensation methods. Compared with the curve fitting method, the error compensation reduced the fitting residual about 18.12–32.64% and improved the correlation about 0.86–1.82%. Then, the calibration curve between the intensity and concentration of the Cu was established. It can be seen that the error compensation method exhibits a higher linear correlation between the intensity and concentration of Cu, which can be applied to the decomposition and correction of overlapping peaks in the LIBS spectrum. © 2017 Optical Society of America

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1. INTRODUCTION

Laser induced breakdown spectroscopy (LIBS) uses the laser point on the sample surface to generate plasma through the plasma spectrum to detect the material composition of the sample [1]. LIBS technology enables rapid and effective detection of solid [2], liquid [3], and gas [4] samples and obtains the material composition information of the sample, thus, achieving multi-element analysis. Compared with the traditional spectral detection methods, such as atomic absorption spectroscopy and x-ray fluorescence spectroscopy, LIBS technology has the advantages of simple operation, low cost, and micro-damage detection [5,6], which can be widely used in food [7], agriculture [8], biomedicine [9], geological survey [10], and other fields. The qualitative analysis and quantitative analysis are two types of analysis methods of the LIBS spectrum. It is the prerequisite and basis for qualitative [11] and quantitative [12] analysis to extract the characteristic spectrum accurately and effectively from the original LIBS spectrum.

In the collection process of LIBS, due to the nature of the element line and the influence of the experimental equipment and environment, the spectral lines in the LIBS spectrum inevitably exist with overlapping interference. Therefore, it is necessary to decompose the overlapping peaks and extract the characteristic spectral lines. The traditional method treats the spectral peaks as a Lorentz [13] or Voigt [14] profile, and then the simulation spectrum is obtained using a curve fitting method. The least squares curve fitting algorithm has been used by some researchers in the analysis of overlapping peaks of the LIBS spectrum [15–18]. The main idea of this algorithm is to consider the overlapping peaks as superposition of two Lorentz functions and obtain the vital parameters of the Lorentz function by least squares fitting, such as half-width, center wavelength, and spectra intensity. Wang [15] *et al.* implemented Lorentz curve fitting for LIBS lines and acquired the peak area, half-width, and peak height. This algorithm does not take into account the overlap interference between the lines. Yao [16] *et al.* applied Lorentz curve fitting to investigate

the decomposition and correction analysis of C–Fe overlapping peaks and the Fe characteristic spectrum was obtained, which was used to a calibration experiment for Fe in the fly ash. Zhang [17] *et al.* used the fractional differential theory combined with Lorentz fitting to investigate the decomposition and correction of the overlapping peaks. Compared with the traditional curve fitting method, this theory improved the accuracy of the decomposition process. Wang [18] *et al.* proposed a damping least squares algorithm for the separation and extraction of overlapping peaks in LIBS. The damping factor was added to the iterative step, and the iterative step was dynamically adjusted to effectively improve the fast convergence of the iteration according to the feedback information from each iteration. In the calibration experiment of Fe–Cr, the damping least squares algorithm exhibited better results than the traditional least squares algorithm.

These curve fitting algorithms can achieve the decomposition of overlapping peaks very well, but these algorithms do not take into account the fitting residuals that represent partial spectral information. That is, the fitting spectrum probably loses partial spectral information without considering the fitting residuals. This paper investigates the decomposition and correction of overlapping peaks using an error compensation algorithm. The error is fed back into the overlapping peaks multiple times and performing multiple fitting processes to obtain a result with a lower fitting residual. In order to verify the validity of the method, the decomposition and correction of Cu–Fe overlapping peaks and the quantitative experiment of Cu were done. Compared with the traditional curve fitting method, the experiment results show that the error compensation method has a good effect on the decomposition and correction of overlapping peaks in the spectrum.

2. EXPERIMENTAL METHOD

For the overlapping peaks formed by the interference between spectral lines, the overlapping lines satisfy the principle of linear superposition. The relationship between spectral intensity and wavelength is expressed as Eq. (1),

$$I_\lambda = I_{1\lambda} + I_{2\lambda} = \frac{i_1 w_1^2}{(\lambda - \lambda_1)^2 + w_1^2} + \frac{i_2 w_2^2}{(\lambda - \lambda_2)^2 + w_2^2}, \quad (1)$$

where I_λ is the spectral intensity of overlapping peaks, $I_{\lambda 1}$ and $I_{\lambda 2}$ are the spectral intensity of two spectral lines, λ is the wavelength information of the overlapping peaks, λ_1 , i_1 , w_1 and λ_2 , i_2 , w_2 are central wavelength, intensity, and half-width of two spectral lines, respectively, which are three important parameters to be fitted in curve fitting.

The main steps of the traditional curve fitting method include: the three parameters of the two peaks are obtained by Lorentz fitting; the parameters and the overlapping region are introduced into the Lorentz function to obtain the two simulation spectral lines, where the simulation overlapping peaks can also be obtained by linear superposition of the two spectral lines; and the decomposition and correction of overlapping peaks are realized.

Compared with the Lorentz function, the Voigt function, which is the linear superposition of the Lorentz function and the Gauss function, can better represent the profile

characteristics of spectral lines. The relation between the spectral intensity and the wavelength is expressed as Eq. (2),

$$I_\lambda = I_{1\lambda} + I_{2\lambda} = \left[a_1 \frac{i_1 w_1^2}{(\lambda - \lambda_1)^2 + w_1^2} + (1 - a_1) i_1 e^{-\frac{(\lambda - \lambda_1)^2}{2w_1^2}} \right] + \left[a_2 \frac{i_2 w_2^2}{(\lambda - \lambda_2)^2 + w_2^2} + (1 - a_2) i_2 e^{-\frac{(\lambda - \lambda_2)^2}{2w_2^2}} \right], \quad (2)$$

where a_1 and a_2 are constant values; the a_1 , a_2 , and other parameters of the spectral peaks are also obtained by the curve fitting method. Thus, decomposition and correction of overlapping peaks is achieved.

In order to reduce the residual of the traditional decomposition process, the residual information is preferably fed back to the fitting spectrum. An error compensation method was investigated to implement the decomposition and correction of overlapping peaks in this study. The main idea of this method is that the residual is fed back to the overlapping spectral lines, then, the fitting calculation is repeated for several times to reduce the residual. Considering that the fitting residual values of each data point obtained after the curve fitting is very close, a linear regression for residual was done to obtain the estimated value of the residual, and then the estimated value of the residual was fed back to the two spectral lines. The process of this method is described as follows:

1. The original overlapping peaks I_λ were fitted using the Voigt function to obtain the two spectral lines and the fitting residual. Assuming that the decomposition results of the two spectral lines are P_1 , P_2 , the intensity values are $I_{\lambda 1}$ and $I_{\lambda 2}$, the fitting residual is Sr_λ , and the acquired $I_\lambda = I_{1\lambda} + I_{2\lambda} + Sr_\lambda$.

2. Considering that the residual value of each point is roughly equal to each other, the residual value Sr_λ is used for linear regression, which is the result acquired as Eq. (3),

$$Sr_\lambda = Sr1_\lambda + err_\lambda, \quad (3)$$

where $Sr1_\lambda$ is the regression result, and err_λ is the error after linear regression.

3. The $Sr1_\lambda$ will be compensated to the spectral peaks P_1 and P_2 , respectively. The intensity values of each point of the two peaks are updated as Eq. (4):

$$I_{1\lambda} = I_{1\lambda} + \frac{I_{1\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1_\lambda; \quad I_{2\lambda} = I_{2\lambda} + \frac{I_{2\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1_\lambda. \quad (4)$$

4. The peaks P_1 and P_2 are linearly superimposed to obtain P_λ after the update of the intensity values,

$$P_\lambda = I_{1\lambda} + I_{2\lambda}. \quad (5)$$

5. The P_λ will be also fitted based on the Voigt function with the fitting result obtained as Eq. (6),

$$P_\lambda = P_{1\lambda} + P_{2\lambda} + S_{p\lambda}, \quad (6)$$

where $S_{p\lambda}$ represents the fitting error. Compared with the original overlapping peaks I_λ , a new residual Sr'_λ is obtained by Eq. (7):

$$Sr'_\lambda = I_\lambda - (P_{1\lambda} + P_{2\lambda}). \quad (7)$$

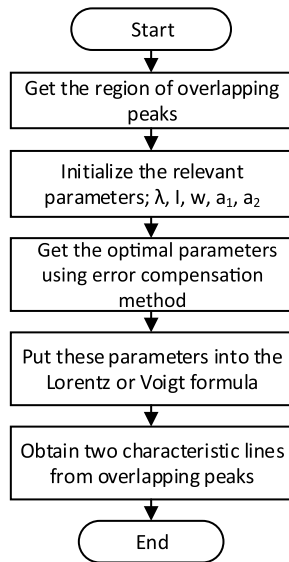


Fig. 1. Error compensation for decomposition of overlapping peaks.

6. The residual is used for linear regression and the acquired $Sr'_\lambda = Sr1'_\lambda + err'_\lambda$. Then, the intensity value of the peaks P_1 and P_2 are updated again, according to $Sr1'_\lambda$:

$$I_{1\lambda} = I_{1\lambda} + \frac{I_{1\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1_\lambda + \frac{I_{1\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1'_\lambda,$$

$$I_{2\lambda} = I_{2\lambda} + \frac{I_{2\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1_\lambda + \frac{I_{2\lambda}}{I_{1\lambda} + I_{2\lambda}} Sr1'_\lambda. \quad (8)$$

7. Repeat the curve fitting and error compensation process as mentioned above until satisfying a given number of repetition or error values that are less than a given threshold or they no longer change. The flow chart of the main correction step of overlapping peaks is shown in Fig. 1.

The total fitting residual and correlation are two common criteria for judging the results of curve fitting. Suppose that there are two vectors including $y = (y_1, y_2 \dots y_n)$ and $k = (k_1, k_2 \dots k_n)$, which represent experiment data and fitting data. The total fitting residual is expressed as

$$E = \sum_{i=1}^n |y_i - k_i|. \quad (9)$$

The correlation is expressed as

$$R = \frac{\sum_{i=1}^n [(y_i - \bar{y})(k_i - \bar{k})]}{\left[\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (k_i - \bar{k})^2 \right]^{1/2}}, \quad (10)$$

where the $\bar{y}$ and $\bar{k}$ are the average values of vector y and vector k , E and R are the total fitting residual and the correlation between vector y and vector k .

3. EXPERIMENT AND DISCUSSIONS

A. Experimental Device and Samples

Spectral data were obtained using a LIBS manufactured by the company IVEA. The spectroscopy mainly includes a laser analysis head, spectrometer, and detector. The laser analysis head

consists of a spot shaping system, a signal collection system, and a sample imaging system. The plasma is generated by a neodymium-doped yttrium aluminum garnet (Nd:YAG) laser (wavelength is 1064 nm, pulse energy is 35 mJ, laser frequency is 1 Hz). A built-in camera imaging device can achieve real-time observation and position of samples by the spot and a three-dimensional fine-tuning rack (adjusted within 50 mm in the x , y , and z directions). The plasma optical radiation signal was collected by a spectrometer (spectral range is 200–1000 nm, resolution is about 0.1–0.2 nm). The detector is an intensified charge-coupled device (ICCD) and can be synchronized with the laser in the external trigger mode.

The original sample was a $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ crystal with a mass fraction of 99.98%. The crystals were configured to 0.01, 0.05, 0.1, 0.2, and 0.3 mol/L solutions [19,20]. Different concentrations of the $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution were added to the Ni alloy's surface for quantitative analysis of Cu. The matrix elements in Ni alloys have no effect on the experimental results, and the Ni alloys were used as samples for measurement after natural air drying.

B. Experimental Results

In the quantitative experiment, 0.01, 0.05, 0.1, 0.2, and 0.3 mol/L LIBS spectrums of the Cu element were obtained. The best experimental parameters (laser energy 35 mJ, time delay $-7.5 \mu\text{s}$, gate width 5 ms) were determined by the relevant pre-experiments, where the same measurement of the sample was repeated 10 times and the average was taken as the final spectrum. As shown in Fig. 2(a), the spectrums

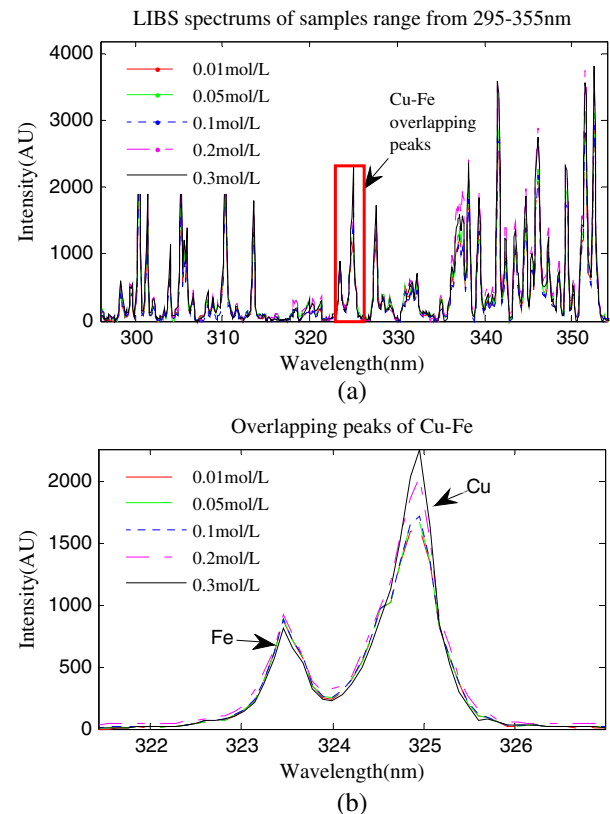


Fig. 2. Cu element LIBS spectrum and the Cu-Fe overlapping peaks.

Table 1. Fitting Parameters Acquired by Different Methods

Parameters	Lorentz	Voigt	Compensation
λ_1 (nm)	323.4805	323.5063	323.5102
I_1 (AU)	0.2564	0.5255	0.5277
w_1 (nm)	0.1912	0.0755	0.0758
λ_2 (nm)	324.8794	324.9029	324.9013
I_2 (AU)	0.5239	1.1479	1.1486
w_2 (nm)	0.2928	0.0816	0.0816
a_1		2.9282	2.9346
a_2		3.5021	3.4996

ranging from 295 to 355 nm were obtained, and the Cu–Fe overlapping peaks are marked by a red rectangle. In order to see clearly, the spectra marked by a red rectangle were fractionated gained.

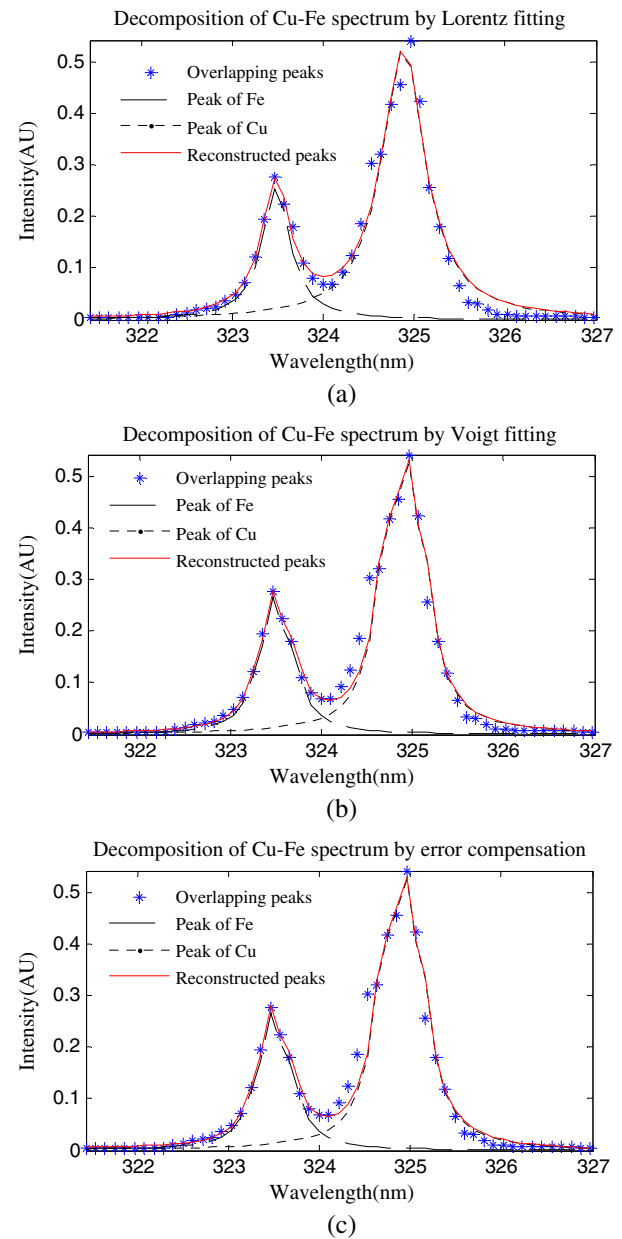
Figure 2(b) shows the spectra lines collected from five different concentrations of Cu. The overlapping interference has occurred between the spectral line of Cu element Cu I: 324.754 nm and the spectral line of Fe element Fe I: 323.461 nm.

The overlapping lines are decomposed and corrected by the curve fitting method and the error compensation method, respectively (take the spectrum of 0.01 mol/L as an example). Table 1 shows the fitting parameters acquired by the curve fitting method and the error compensation method.

Figures 3(a)–3(c) show the decomposition results of the Cu–Fe overlapping spectrum by the curve fitting and the error compensation method, and the spectra lines of the Cu element and Fe element are obtained. Both methods are effective in extracting the two spectral lines from the overlapping peaks. The final results show that after reconstruction (linear superposition) of the two spectral lines, the results obtained by the error compensation method have good similarity with the original overlapping peaks and improve the accuracy of the decomposition results. The decomposition and correction results of Cu–Fe overlapping peaks, using different methods in the LIBS spectrum, are shown in Table 2.

In Table 2, E represents the fitting residual, and C is the correlation coefficient. For the curve fitting method, Voigt fitting exhibits a better decomposition result than that of Lorentz fitting. Compared with Lorentz fitting, the error compensation method can reduce the error obtained by the curve fitting method and increase the correlation coefficient between the reconstructed spectrum and the original overlapping peaks. The experimental results show that the error compensation method can be applied to the overlapping peaks' correction to improve the accuracy of the decomposition and correction results of overlapping peaks.

In order to verify the feasibility of the error compensation method, the calibration experiment of Cu was done. The Cu spectral lines of five different concentrations were obtained to realize the calibration experiment. As shown in Fig. 4(a), the calibration result was obtained between the concentration and intensity of the Cu element before the overlapping peaks correction. Due to the overlapping interference between the

**Fig. 3.** Decomposition results acquired by Lorentz fitting, Voigt fitting, and error compensation (with overlap interference of the Fe element).

Cu spectral line and the Fe spectral line, the calibration result was not too ideal, and the linear correlation is 0.9763.

Quantitative analysis of Cu was carried out after decomposition and correction of the overlapping lines by the curve fitting method and the error compensation method; the results are shown in Figs. 4(b)–4(d). It can be seen that after decomposition and correction of the overlapping peaks, the calibration result of Cu was effectively improved. Figures 4(b) and 4(c) show that the Voigt fitting method exhibits better quantitative result than the Lorentz fitting method. From Fig. 4(d), the correlation coefficient of the calibration curve between the concentration and intensity of Cu is 0.9956 by the

Table 2. Comparison of Decomposition Results Between the Curve Fitting Method and the Error Compensation Method

Cu-Fe Overlapping Peaks	Lorentz Fitting	Voigt Fitting	Error Compensation
0.01 mol/L	$E = 0.7878$, $C = 0.9697$	$E = 0.6166$, $C = 0.9766$	$E = 0.5969$, $C = 0.9848$
0.05 mol/L	$E = 0.6992$, $C = 0.9832$	$E = 0.5285$, $C = 0.9832$	$E = 0.4592$, $C = 0.9914$
0.1 mol/L	$E = 0.9423$, $C = 0.9762$	$E = 0.7472$, $C = 0.9806$	$E = 0.7065$, $C = 0.9889$
0.2 mol/L	$E = 1.2645$, $C = 0.9718$	$E = 0.9646$, $C = 0.9787$	$E = 0.9241$, $C = 0.9870$
0.3 mol/L	$E = 1.1436$, $C = 0.9712$	$E = 0.9091$, $C = 0.9808$	$E = 0.8627$, $C = 0.9891$

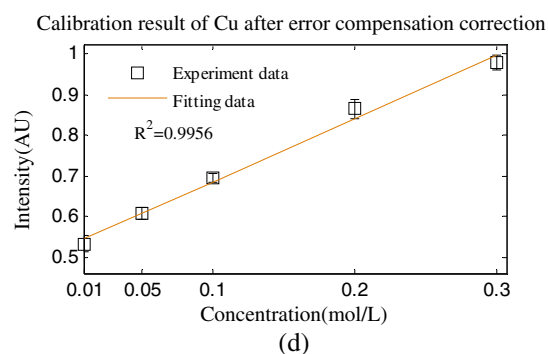
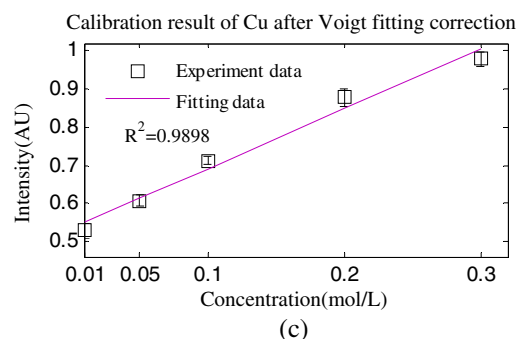
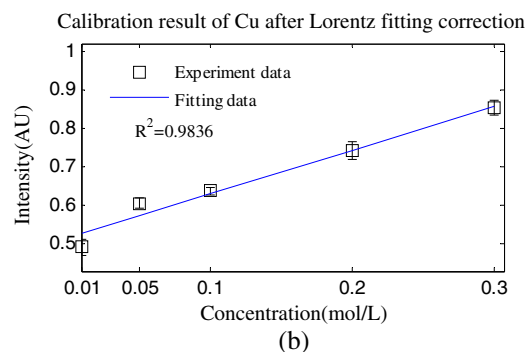
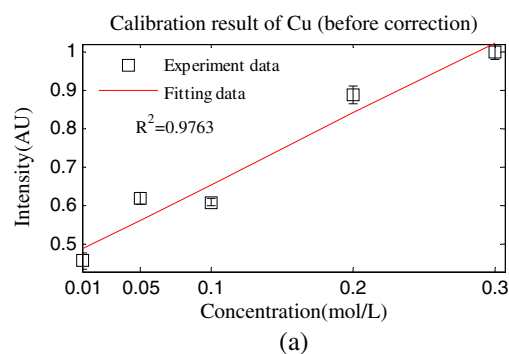
error compensation method, which is better than the Voigt fitting method. The error compensation exhibits a good decomposition result, and the intensity of Cu spectral lines has a good linear correlation with the concentration of Cu.

C. Verification Results

Figure 5 shows the LIBS spectrum of Longjing green tea (the spectrum data were provided by the AnHui Institute of Optics and Fine Mechanics) in the range of 256–263.5 nm, and there are three kinds of overlapping spectral lines (as shown in Fig. 5, overlapping regions 1, 2, and 3 range from 259 to 261.5 nm). The overlapping peaks 1, 2, and 3 are common interference between the spectral lines in the LIBS spectrum. They were decomposed and corrected by the Lorentz curve fitting method and error compensation method, respectively. The decomposition results using the curve fitting method and error compensation method of the overlapping peaks are shown in Fig. 6. The correlation and the fitting residuals between decomposition results and original overlapping peaks are shown in Table 3.

Figures 6(b), 6(d), and 6(f) are the decomposition results using the Lorentz fitting method of the overlapping peaks 1, 2, and 3, while Figs. 6(a), 6(c), and 6(e) are the decomposition results using the error compensation method. Both methods are effective in extracting the two spectral lines from the overlapping peaks. The final results show that after reconstruction (linear superposition) of the two spectral lines, the results obtained by the error compensation method have good similarity with the original overlapping peaks, and it improved the accuracy of the decomposition results. The error and correlation acquired by the two methods are shown in Table 3.

In the experiment, the Cu-Fe overlapping peaks in the LIBS spectrum were decomposed and corrected by the curve fitting method and the error compensation method, respectively. The decomposition results prove that the Voigt fitting method obtained a better correction effect than the Lorentz fitting method. Compared with the curve fitting method, the error compensation method exhibits a better decomposition result and obtains a small residual and large correlation. In the quantitative analysis of Cu, when the calibration analysis between the intensity and concentration was done before decomposition

**Fig. 4.** Calibration results of Cu before and after overlapping correction by curve fitting and the error compensation method.

and correction of the Cu-Fe overlapping peaks, the calibration results have a big error, and the correlation coefficient is low. The accuracy of the calibration results can be improved effectively when the overlapping lines are decomposed and corrected, and the error compensation method exhibits a high linear correlation. The decomposition and correction results

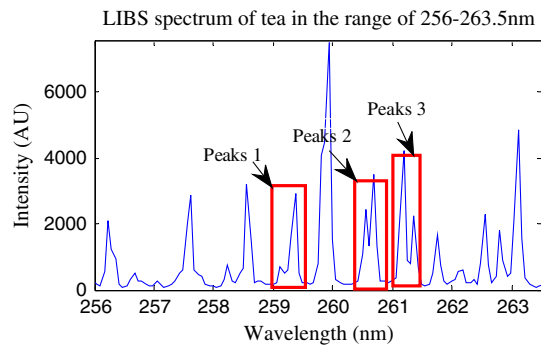


Fig. 5. LIBS spectrum of tea in the range of 254–264 nm.

Table 3. Error and Correlation Acquired After Correction of Overlapping Peaks by the Curve Fitting Method and the Error Compensation Method

Peaks	Lorentz Fitting	Error Compensation
1	$E = 0.1744$, $C = 0.9981$	$E = 0.1353$, $C = 0.9987$
2	$E = 0.2323$, $C = 0.9877$	$E = 0.0938$, $C = 0.9997$
3	$E = 0.1215$, $C = 0.9888$	$E = 0.0509$, $C = 0.9998$

of overlapping peaks 1, 2, and 3 in tea in the LIBS spectrum can be seen in Table 3. The error compensation method can reduce the error obtained by the curve fitting method and increase the correlation coefficient between the reconstructed spectrum and the original overlapping peaks.

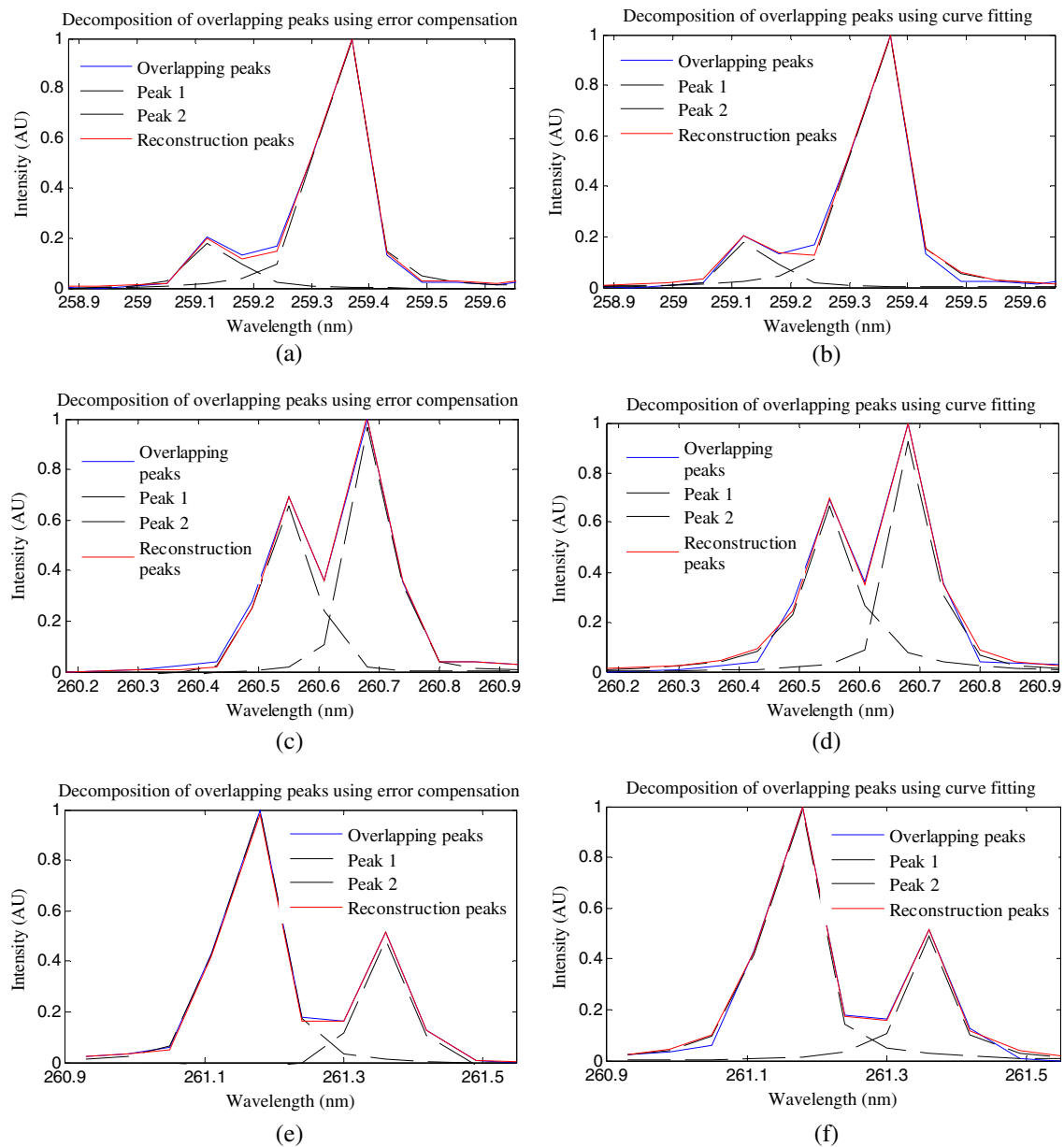


Fig. 6. Decomposition and correction results of overlapping peaks by Lorentz curve fitting and error compensation.

4. CONCLUSION

From the above experiment, the results proved that the error compensation method is reliable for the decomposition and correction of overlapping peaks. Compared with the curve fitting method, it is possible to reduce the fitting error about one half, and the correlation also has a small increase. In the quantitative analysis of Cu, there is a good linear correlation between the concentration and intensity of the Cu after the decomposition and correction of the Cu–Fe overlapping peaks. The error compensation method effectively improved the accuracy of the results in the decomposition of overlapping peaks and the extraction of spectral lines, which provides a reliable data processing method for the study of spectral lines in the qualitative and quantitative analysis.

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